The Journal of Open Source Software

CÆLUM: An academic High-Order Finite-Volume solver for the compressible Euler equations and related models

Adrian Navas-Montilla [1*] **and Isabel Echeverribar** [2*]

1 Fluid Dynamics Technologies, i3A-University of Zaragoza, Spain. **2** Instituto Tecnológico de Aragón – ITA, Zaragoza, Spain. * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Philip Cardiff

Reviewers:

- @iBatistic
- @nasserma

Submitted: 09 October 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

In the last decade, very high order numerical methods have become very popular within the Computational Fluid Dynamics (CFD) and Numerical Weather Prediction (NWP) communities. Traditional low order –i.e. equal or lower than 2– methods have been widely adopted by industry and academia to simulate fluid flows, but they show important limitations as they feature high diffusion and dispersion errors which may lead to non-physical solutions. On the other hand, high order schemes are able to provide accurate solutions, with small diffusion and dispersion errors, and feature a high computational efficiency (Ferrer et al., 2023; Wang et al., 2013). They are well suited for multi-scale problems, e.g. turbulent flows, flows involving sharp gradients such as shocks, acoustics, etc. High order solvers have gained attention due to their notable performance when running on modern High-Performance Computing (HPC) architectures.

CÆLUM is a hydrodynamic solver for compressible flows, which solves the Euler equations of gas dynamics and other related models by means of very high order essentially non-oscillatory finite volume schemes in cartesian meshes. The core of the solver is written in C but pre-processing and post-processing can be easily done using Jupyter Notebook as provided in the examples. The Weighted Essentially Non-Oscillatory (WENO) (Jiang & Shu, 1996) and Targeted Essentially Non-Oscillatory (TENO) (Fu, 2019b) spatial reconstructions, in combination with Runge-Kutta integrators (Gottlieb et al., 2001), are used in CÆLUM to achieve a high resolution in smooth regions of the flow and to capture discontinuous solutions without spurious oscillations. These schemes also allow to adopt an Implicit Large Eddy Simulation (iLES) framework when computing turbulent flows (Grinstein et al., 2007; San & Kara, 2015; Solán-Fustero et al., 2021). The numerical diffusion plays the role of the explicit dissipative sub-grid model, being this a favorable choice when seeking simplicity and computational efficiency. In summary, this hydrodynamic solver can compute turbulent compressible flows, as well as low Mach number flows such as simplified dry atmospheric flows and other wave propagation phenomena.

Statement of need

Given the limitations of traditional low-order numerical methods mentioned above, there is a critical need for advanced, high-order computational tools. The increasing complexity of engineering and scientific problems motivates a shift toward methods that can deliver both high accuracy and computational efficiency. Therefore, it is important to popularize such methods and make them available for the scientific community, in particular, for students and novel researchers who are often limited by the accessibility of these tools.

CÆLUM is designed to be used both by researchers and by students in the field of CFD, atmospheric flows and related disciplines. In fact, the simplicity and compactness of the code make it particularly suitable for those taking their first steps in the numerical simulation of flows using high-order methods. It is designed from an academic perspective, where clarity and accessibility are prioritized. Therefore, it includes user-friendly pre-processing and post-processing tools based on Python and Jupyter Notebook. The repository comes with a series of Python scripts for the configuration and visualization of various example flows, ranging from simple scalar advection in 1D to more complex atmospheric or compressible flow cases in 3D. These scripts rely on standard libraries such as `numpy`, `matplotlib` and `pyvista`, the latter being a powerful module for data visualization and rendering. Additionally, several Jupyter Notebooks are included, where the steps for configuring the simulation tool and visualizing the results are explained in detail.

A key feature of this software is that it also allows the simulation of simplified dry atmospheric flows in the meso- and micro-scale. For this, CÆLUM uses the compressible Euler equations with gravity source term, composed by the equations for the conservation of mass, momentum and energy (Ghosh & Constantinescu, 2016). This approach offers several benefits for the selected application, such as the ability to conserve mass and energy with machine accuracy when using a suitable discretization. Another benefit of this approach is that many numerical advances developed by the CFD community can be easily adapted (Giraldo & Restelli, 2008).

Regarding numerics, the simulation code uses WENO (Jiang & Shu, 1996), TENO (Fu, 2019b), and optimal polynomial reconstructions on Cartesian meshes. For simplicity, a discretization strategy based on the midpoint rule that employs independent 1D reconstructions in each Cartesian direction is used (Zhang et al., 2011). This approach prevents from performing multi-dimensional reconstructions and Gaussian integration at the cell faces, thereby drastically reducing the complexity of the algorithms as well as computational expenses. This is done at the cost of not achieving a genuinely high order of accuracy, which is not critical when computing shocked problems, underresolved flows, or flows with discontinuities and sharp gradients (Fu, 2019a; San & Kara, 2015; Zhang et al., 2011). As a result, we provide a simple and versatile computational code that can be applied to a wide variety of problems and enables iLES.

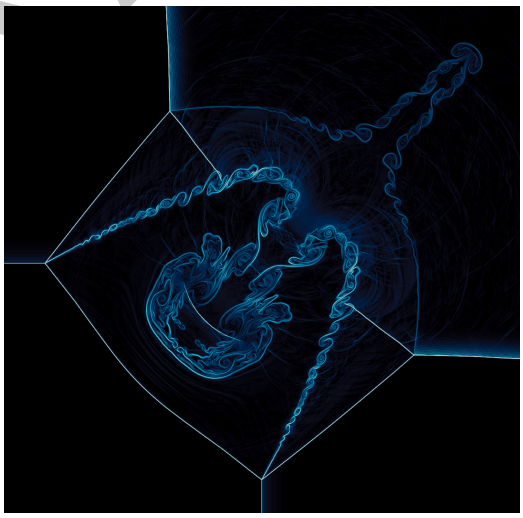

Figure 1: Density gradient (schlieren-like) representation of a 2D Riemann Problem from Lax & Liu (1998). Solution computed by a 7-th order WENO scheme.

As an example, Figure 1 displays the solution provided by CÆLUM of a 2D Riemann Problem from Lax & Liu (1998), which is a typical benchmark for this type of simulation codes. The figure shows the density gradient representation, which allows to reveal the main features of

DRAFT

the flow. Shock waves as well as slip –i.e. contact– lines where Kelvin-Helmholtz vortices form are accurately captured.

Regarding the implementation of the solver, CÆLUM uses OpenMP, a directive-based threading library that allows parallel computing in multi-core CPUs. The modification of the base code by the inlcusion of pragmas is minimal, preserving the readability and clarity of the code.

We can find other open-source simulation codes using high order non-oscillatory finite volume schemes for the more-general Navier-Stokes equations, aimed at similar aplications, which show a superior performance than most state-of-the-art comercial codes. Some examples are OpenSBLI (Lusher et al., 2021), HTR Solver (Di Renzo et al., 2020), JAX-Fluids (Bezgin et al., 2023), UCNS3D solver (Antoniadis et al., 2022), URANOS (De Vanna et al., 2023), STREAmS (Bernardini et al., 2021) and HyPar (Ghosh, 2015). Analogously, there are other open-source solvers based on the Discontinuous Galerkin method such as HORSES3D (Ferrer et al., 2023), GALÆEXI (Kempf et al., 2024), Nek5000 (Markidis et al., 2015) and NebulaSEM (Abdi, 2024). We can also find finite volume solvers aimed exclusively at atmospheric applications, such as the OpenFoam-based GEA (Girfoglio et al., 2023) and a WENO-based model for the moist atmosphere (Norman et al., 2023). Note that CÆLUM represents an alternative for those seeking a simpler and more academic tool, as it is designed to be pedagogical and versatile in terms of applications, though it has some limitations when considering realistic cases. It can also serve as a testbed for evaluating novel Riemann solvers and spatial reconstruction methods.

The model equations

We herein consider the compressible Euler equations with gravitational source term, given by

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{Continuity}$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + p\mathbf{I}) = \rho \mathbf{g} \tag{Momentum}$$

$$\frac{\partial E}{\partial t} + \nabla \cdot ((E + p)\mathbf{v}) = \rho \mathbf{v} \cdot \mathbf{g} \tag{Energy}$$

where ρ is density, $\mathbf{v}$ is the velocity vector, p is pressure, $\mathbf{g} = (0, 0, -g)^T$ is the specific gravity force and $\mathbf{I}$ is the identity matrix. The energy is defined as the sum of kinetic and internal energy $E = \rho(\frac{1}{2}\mathbf{v} \cdot \mathbf{v} + e)$.

Additionally, scalar transport can also be considered as the following equation is implemented

$$\frac{\partial u}{\partial t} + \nabla \cdot (\mathbf{v} u) = 0$$

where u is the transported quantity and $\mathbf{v}$ is the advection velocity. It is also possible to compute the Burgers equation when setting $\mathbf{v} = 1/2(u, u, u)^T$.

Further details on the model equations and numerical resolution methods can be found in (Navas-Montilla et al., 2024; Navas-Montilla & Echeverribar, 2023).

Previous work using the software

CÆLUM was developed and first used in Navas-Montilla & Echeverribar (2023) presenting novel methodology to construct very high order well-balanced schemes for the computation of the Euler equations with gravitational source term, with application to NWP. The objective of this work was twofold: first, to assess the use of augmented solvers in computing the Euler equations with gravity, and second, to evaluate the performance of the novel TENO reconstruction for NWP. Afterwards, a more traditional implementation of the Euler equations with gravity

based on the use of fluctuation variables was implemented in different forms to also allow for the conservation of total energy. The resulting schemes were evaluated in terms of their spectral resolution for the computation of turbulent flows in the dry atmosphere. The results were presented in Navas-Montilla et al. (2024), with the objective of determining whether or not these models can be used to build an iLES framework, shedding light on their potential advantages or limitations in representing under-resolved atmospheric processes in the meso- and micro-scales.

Example cases

The example cases provided in the repository showcase a diverse suite of test problems that are commonly used to validate and demonstrate the capabilities of computational fluid dynamics codes. These include scalar problems like linear advection and Burgers' equation in 1D, 2D and 3D, compressible flow simulations such as Riemann problems (single- and multi-component flow), shock–bubble interactions, and a Richtmyer–Meshkov instability for turbulent mixing. Additionally, atmospheric flow cases feature colliding thermal bubbles and descending thermal plumes over complex terrain. Each case is supported by Python scripts (or Jupyter notebooks) and corresponding visual animations, making them useful for verifying the numerical solvers and exploring the underlying physical phenomena. Apart from this, the code also provides an automated testing script that performs a convergence rate test, executes four Riemann problems, and runs an atmospheric flow benchmark.

Acknowledgements

Part of this work has been funded by the Spanish Ministry of Science, Innovation and Universities - Agencia Estatal de Investigación (10.13039/501100011033) and FEDER-EU under project-nr. PID2022-141051NA-I00.

References

Abdi, D. S. (2024). NebulaSEM: A high-order discontinuous galerkin spectral element code for atmospheric modeling. *Journal of Open Source Software*, *9*(96), 6448. https://doi.org/10.21105/joss.06448

Antoniadis, A. F., Drikakis, D., Farmakis, P. S., Fu, L., Kokkinakis, I., Nogueira, X., Silva, P. A. S. F., Skote, M., Titarev, V., & Tsoutsanis, P. (2022). UCNS3D: An open-source high-order finite-volume unstructured CFD solver. *Computer Physics Communications*, *279*, 108453. https://doi.org/10.1016/j.cpc.2022.108453

Bernardini, M., Modesti, D., Salvadore, F., & Pirozzoli, S. (2021). STREAmS: A high-fidelity accelerated solver for direct numerical simulation of compressible turbulent flows. *Computer Physics Communications*, *263*, 107906. https://doi.org/10.1016/j.cpc.2021.107906

Bezgin, D. A., Buhendwa, A. B., & Adams, N. A. (2023). JAX-fluids: A fully-differentiable high-order computational fluid dynamics solver for compressible two-phase flows. *Computer Physics Communications*, *282*, 108527. https://doi.org/10.1016/j.cpc.2022.108527

De Vanna, F., Avanzi, F., Cogo, M., Sandrin, S., Bettencourt, M., Picano, F., & Benini, E. (2023). URANOS: A GPU accelerated navier-stokes solver for compressible wall-bounded flows. *Computer Physics Communications*, *287*, 108717. https://doi.org/10.1016/j.cpc.2023.108717

Di Renzo, M., Fu, L., & Urzay, J. (2020). HTR solver: An open-source exascale-oriented task-based multi-GPU high-order code for hypersonic aerothermodynamics. *Computer Physics Communications*, *255*, 107262. https://doi.org/10.1016/j.cpc.2020.107262

Ferrer, E., Rubio, G., Ntoukas, G., Laskowski, W., Mariño, O. A., Colombo, S., Mateo-Gabín, A., Marbona, H., Manrique de Lara, F., Huergo, D., Manzanero, J., Rueda-Ramírez, A. M., Kopriva, D. A., & Valero, E. (2023). A high-order discontinuous galerkin solver for flow simulations and multi-physics applications. *Computer Physics Communications*, *287*, 108700. https://doi.org/10.1016/j.cpc.2023.108700

Fu, L. (2019a). A low-dissipation finite-volume method based on a new TENO shock-capturing scheme. *Computer Physics Communications*, *235*, 25–39. https://doi.org/10.1016/j.cpc.2018.10.009

Fu, L. (2019b). A very-high-order TENO scheme for all-speed gas dynamics and turbulence. *Computer Physics Communications*, *244*, 117–131. https://doi.org/10.1016/j.cpc.2019.06.013

Ghosh, D. (2015). *HyPar Repository (https://github.com/debog/hypar)*.

Ghosh, D., & Constantinescu, E. M. (2016). Well-balanced, conservative finite difference algorithm for atmospheric flows. *AIAA Journal*, *54*(4), 1370–1385. https://doi.org/10.2514/1.J054580

Giraldo, F. X., & Restelli, M. (2008). A study of spectral element and discontinuous galerkin methods for the navier–stokes equations in nonhydrostatic mesoscale atmospheric modeling: Equation sets and test cases. *Journal of Computational Physics*, *227*(8), 3849–3877. https://doi.org/10.1016/j.jcp.2007.12.009

Girfoglio, M., Quaini, A., & Rozza, G. (2023). Validation of an OpenFOAM®-based solver for the euler equations with benchmarks for mesoscale atmospheric modeling. *AIP Advances*, *13*(5). https://doi.org/10.1063/5.0147457

Gottlieb, S., Shu, C.-W., & Tadmor, E. (2001). Strong stability-preserving high-order time discretization methods. *SIAM Review*, *43*(1), 89–112. https://doi.org/10.1137/S003614450036757X

Grinstein, F. F., Margolin, L. G., & Rider, W. J. (2007). *Implicit large eddy simulation* (Vol. 10). Cambridge university press Cambridge. https://doi.org/10.1017/CBO9780511618604

Jiang, G.-S., & Shu, C.-W. (1996). Efficient implementation of weighted ENO schemes. *Journal of Computational Physics*, *126*(1), 202–228. https://doi.org/10.1006/jcph.1996.0130

Kempf, D., Kurz, M., Blind, M., Kopper, P., Offenhäuser, P., Schwarz, A., Starr, S., Keim, J., & Beck, A. (2024). GALÆXI: Solving complex compressible flows with high-order discontinuous galerkin methods on accelerator-based systems. *arXiv Preprint arXiv:2404.12703*. https://doi.org/10.48550/arXiv.2404.12703

Lax, P. D., & Liu, X.-D. (1998). Solution of two-dimensional riemann problems of gas dynamics by positive schemes. *SIAM Journal on Scientific Computing*, *19*(2), 319–340. https://doi.org/10.1137/S1064827595291819

Lusher, D. J., Jammy, S. P., & Sandham, N. D. (2021). OpenSBLI: Automated code-generation for heterogeneous computing architectures applied to compressible fluid dynamics on structured grids. *Computer Physics Communications*, *267*, 108063. https://doi.org/10.1016/j.cpc.2021.108063

Markidis, S., Gong, J., Schliephake, M., Laure, E., Hart, A., Henty, D., Heisey, K., & Fischer, P. (2015). OpenACC acceleration of the Nek5000 spectral element code. *The International Journal of High Performance Computing Applications*, *29*(3), 311–319. https://doi.org/10.1177/1094342015576846

Navas-Montilla, A., & Echeverribar, I. (2023). A family of well-balanced WENO and TENO schemes for atmospheric flows. *Journal of Computational Physics*, *489*, 112273. https://doi.org/10.1016/j.jcp.2023.112273

DRAFT

Navas-Montilla, A., Guallart, J., Solán-Fustero, P., & García-Navarro, P. (2024). Exploring the potential of TENO and WENO schemes for simulating under-resolved turbulent flows in the atmosphere using euler equations. *Computers & Fluids*, *280*, 106349. https://doi.org/10.1016/j.compfluid.2024.106349

Norman, M. R., Eldred, C., & Gopalakrishnan Meena, M. (2023). Investigating inherent numerical stabilization for the moist, compressible, non-hydrostatic euler equations on collocated grids. *Journal of Advances in Modeling Earth Systems*, *15*(10), e2023MS003732. https://doi.org/10.1029/2023MS003732

San, O., & Kara, K. (2015). Evaluation of riemann flux solvers for WENO reconstruction schemes: Kelvin–helmholtz instability. *Computers & Fluids*, *117*, 24–41. https://doi.org/10.1016/j.compfluid.2015.04.026

Solán-Fustero, P., Navas-Montilla, A., Ferrer, E., Manzanero, J., & García-Navarro, P. (2021). Application of approximate dispersion-diffusion analyses to under-resolved burgers turbulence using high resolution WENO and UWC schemes. *Journal of Computational Physics*, *435*, 110246. https://doi.org/10.1016/j.jcp.2021.110246

Wang, Z. J., Fidkowski, K., Abgrall, R., Bassi, F., Caraeni, D., Cary, A., Deconinck, H., Hartmann, R., Hillewaert, K., Huynh, H. T., & others. (2013). High-order CFD methods: Current status and perspective. *International Journal for Numerical Methods in Fluids*, *72*(8), 811–845. https://doi.org/10.1002/fld.3767

Zhang, R., Zhang, M., & Shu, C.-W. (2011). On the order of accuracy and numerical performance of two classes of finite volume WENO schemes. *Communications in Computational Physics*, *9*(3), 807–827. https://doi.org/10.4208/cicp.291109.080410s

DRAFT